Matb102

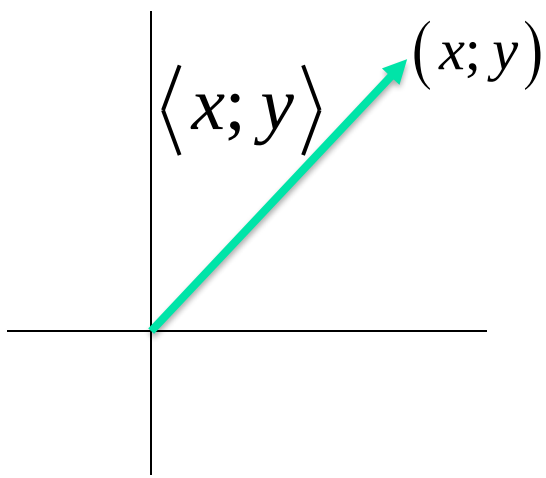
Linear Transformations

Linear Transformation

We use matrix multiplication to transform a vector (point) by either a contraction/ dilation; reflection or rotation.

Linear Transformation

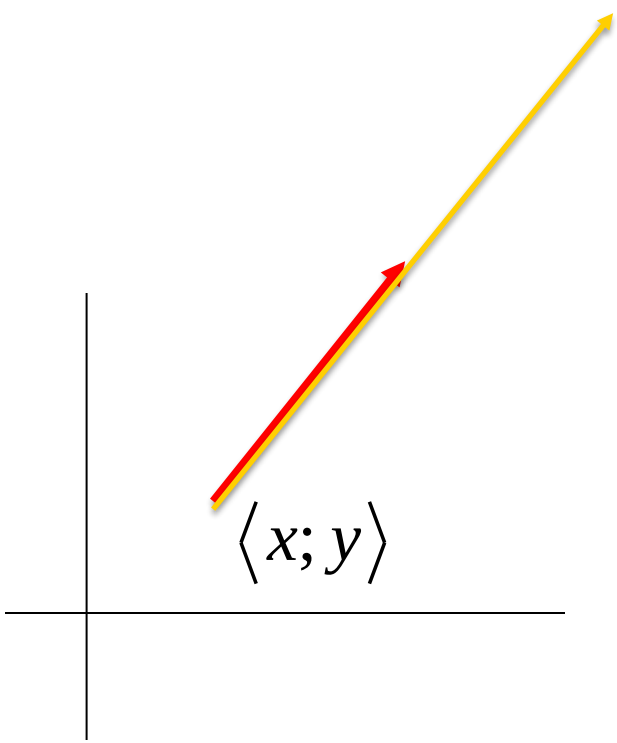
In this section we will consider a point $(x; y)$ and vector $\langle x; y \rangle$ to be the same.



Linear Transformation

Multiply $\langle x; y \rangle$ by a non-negative scalar k

$k \geq 1$ we have a dilation and if $0 \leq k \leq 1$ we have a contraction



$k\langle x; y \rangle = \langle kx; ky \rangle = \langle w_1; w_2 \rangle$ i.e.

$w_1 = kx$

$w_2 = ky$

We can represent these equations as a linear system:

$w_1 = kx + 0y$

$w_2 = 0x + ky$

Linear Transformation

$$w_1 = kx + 0y$$

$$w_2 = 0x + ky$$

we can represent the system of equations in matrix format

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

In $\mathbb{R}^2$ the standard matrix to contract/dilate a vector (point) by a nonnegative factor k is given by the matrix $\begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$.

Linear Transformation

In $\mathbb{R}^3$ the standard matrix to contract/dilate a vector (point) by a nonnegative factor k is given by the matrix $\begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{bmatrix}$.

Dilate $\langle 2;-3;1\rangle$ by 2.

$\langle 2;-3;1\rangle$ by 2

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ -6 \\ 2 \end{bmatrix}$$

$< 4;-6;2 >$

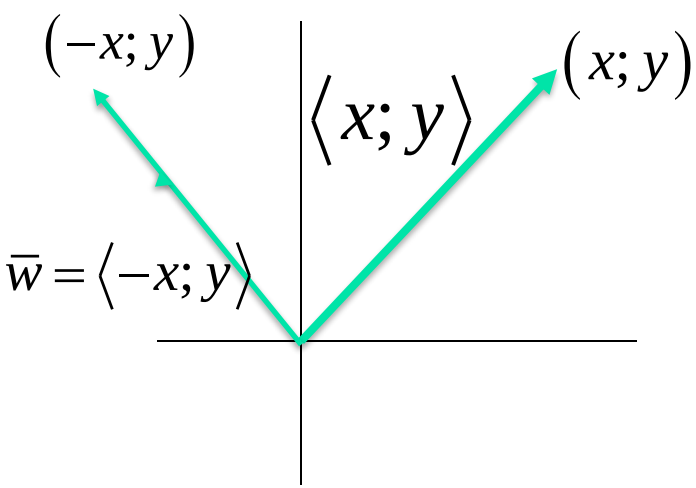
Contract $\langle 2;-3;1\rangle$ by $\frac{1}{3}$.

$$\begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{2}{3} \\ -1 \\ \frac{1}{3} \end{bmatrix} = < \frac{2}{3}; -1; \frac{1}{3} >$$

Reflections

If we reflect $\langle x; y \rangle$ about the y-axis we map the vector into its symmetric image $\langle -x; y \rangle$.

$$\bar{w} = \langle w_1; w_2 \rangle = \langle -x; y \rangle$$



$$\bar{w} = \langle w_1; w_2 \rangle = \langle -x; y \rangle$$

$$w_1 = -1x = -1x + 0y$$

$$w_2 = y = 0x + 1y$$

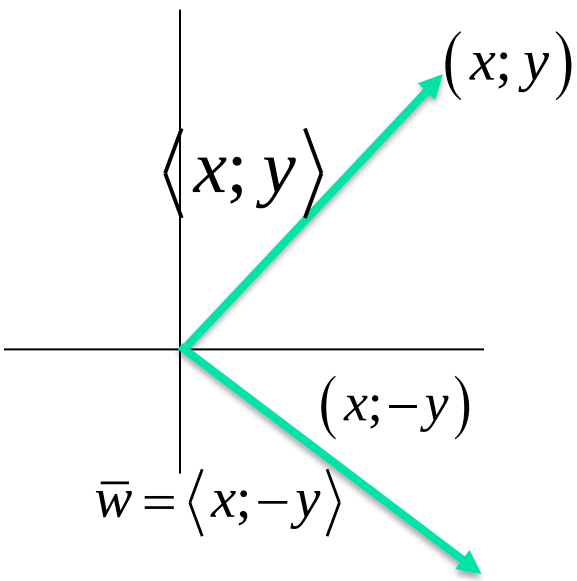
$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Reflections

In $\mathbb{R}^2$ the standard matrix to reflect a vector (point) about the y-axis is $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

Reflections

If we reflect $\langle x; y \rangle$ about the x-axis we map the vector into its symmetric image $\langle x; -y \rangle$.



$$\bar{w} = \langle w_1; w_2 \rangle = \langle x; -y \rangle$$

$$\bar{w} = \langle w_1; w_2 \rangle = \langle x; -y \rangle$$

$$w_1 = 1x = 1x + 0y$$

$$w_2 = -1y = 0x - 1y$$

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Reflections

In $\mathbb{R}^2$ the standard matrix to reflect a vector (point) about the x-axis is $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

Reflections

Reflecting $\langle x; y\rangle$ about the line $y = x$:

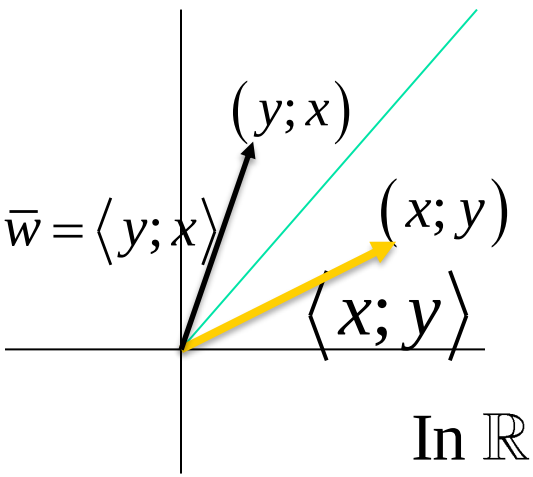
$\langle x; y\rangle$ becomes $\langle y; x\rangle$

$$\bar{w} = \langle w_1; w_2\rangle = \langle y; x\rangle$$

$$w_1 = 1y = 0x + 1y$$

$$w_2 = 1x = 1x + 0y$$

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

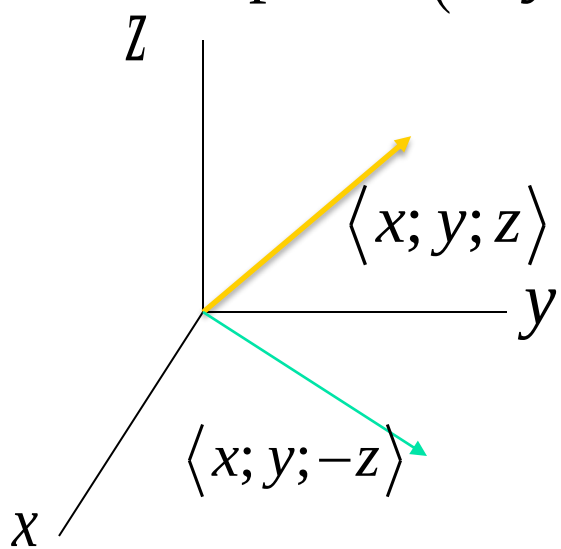


In $\mathbb{R}^2$ the standard matrix to reflect a vector (point) about the line $y = x$ is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

Reflections

Reflections about a plane in $\mathbb{R}^3$:

Reflect a point $(x; y; z)$ about the xy-plane:



$$\bar{w} = \langle w_1; w_2; w_3 \rangle = \langle x; y; -z \rangle$$

$$w_1 = 1x = 1x + 0y + 0z$$

$$w_2 = 1y = 0x + 1y + 0z$$

$$w_3 = -z = 0x + 0y - 1z$$

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Reflections

The transformation matrix to reflect

about the xy-plane $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

The transformation matrix to reflect

about the xz-plane $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

The transformation matrix to reflect

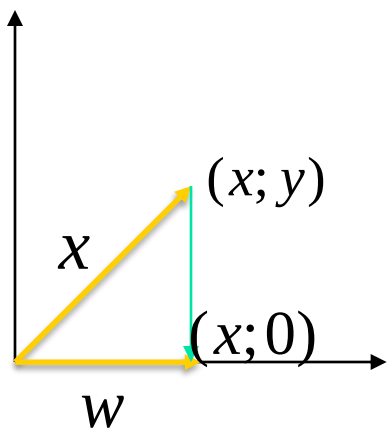
about the yz-plane $\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Orthogonal Projections

An orthogonal projection operator maps a vector into its orthogonal projection onto the x- or y-axis or a plane.

Orthogonal Projections

Orthogonal projection onto the x- axis:



$$\bar{w} = \langle w_1; w_2 \rangle = \langle x; 0 \rangle$$

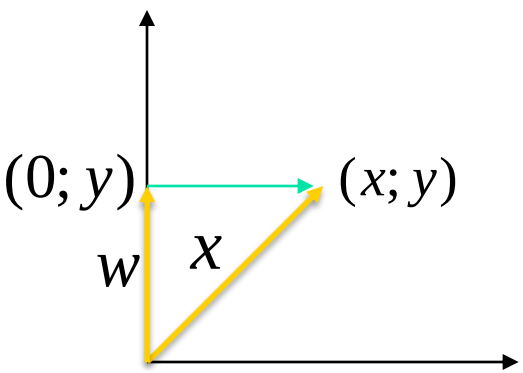
$$w_1 = x = 1x + 0y$$

$$w_2 = 0 = 0x + 0y$$

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Orthogonal Projections

Orthogonal projection onto the y- axis:



$$\bar{w} = \langle w_1; w_2 \rangle = \langle 0; y \rangle$$

$$w_1 = 0 = 0x + 0y$$

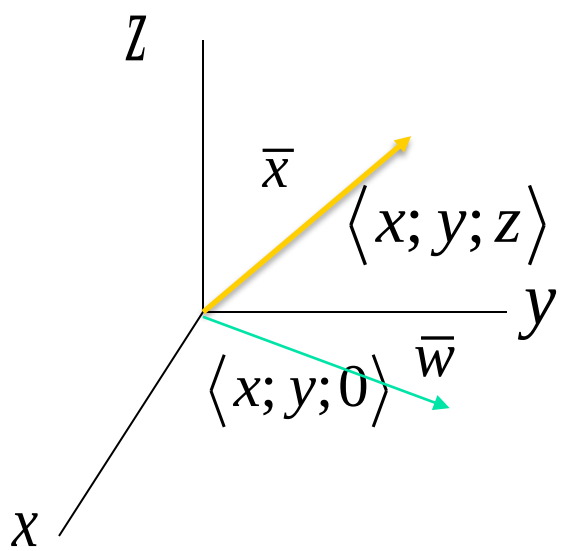
$$w_2 = y = 0x + 1y$$

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Orthogonal Projections

Orthogonal projection onto

the xy- plane:



$$\bar{w} = \langle w_1; w_2; w_3 \rangle = \langle x; y; 0 \rangle$$

$$w_1 = 1x = 1x + 0y + 0z$$

$$w_2 = 1y = 0x + 1y + 0z$$

$$w_3 = 0 = 0x + 0y + 0z$$

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Orthogonal Projections

Transformation matrix for the orthogonal projection onto the *xy*- plane:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Transformation matrix for the orthogonal projection onto the *xz*- plane:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

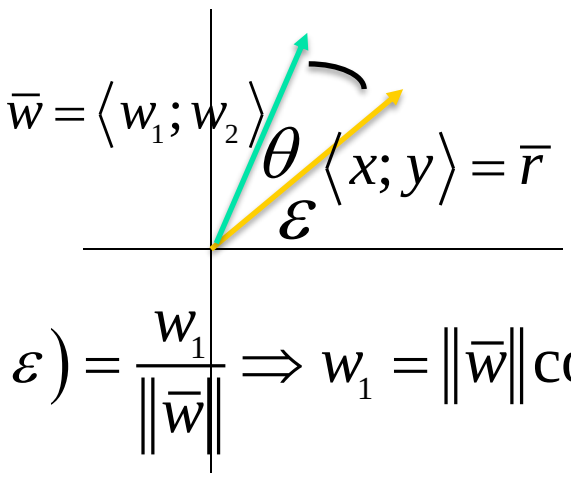
Orthogonal Projections

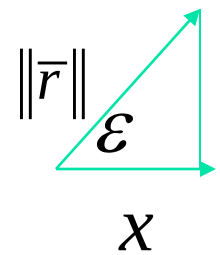
Transformation matrix for the orthogonal projection onto the *yz*- plane:

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotation

A rotation operator will rotate a vector through a fixed positive angle θ.





$$\cos(\theta+\varepsilon) = \frac{w_1}{\|\bar{w}\|} \Rightarrow w_1 = \|\bar{w}\|\cos(\theta+\varepsilon)$$

$$\sin(\theta+\varepsilon) = \frac{w_2}{\|\bar{w}\|} \Rightarrow w_2 = \|\bar{w}\|\sin(\theta+\varepsilon)$$

$$\cos\varepsilon = \frac{x}{\|\bar{r}\|} \Rightarrow x = \|\bar{r}\|\cos\varepsilon$$

$$\sin\varepsilon = \frac{y}{\|\bar{r}\|} \Rightarrow y = \|\bar{r}\|\sin\varepsilon$$

$$x = \|\bar{r}\| \cos \varepsilon \qquad w_1 = \|\bar{w}\| \cos(\theta + \varepsilon)$$

$$y = \|\bar{r}\| \sin \varepsilon \qquad w_2 = \|\bar{w}\| \sin(\theta + \varepsilon)$$

Rotation

$\|\bar{r}\| = \|\bar{w}\|$ (same vector just rotated)

Let r denote the lenght.

$$w_1 = r\cos(\theta + \varepsilon) = r[\cos\theta\cos\varepsilon - \sin\theta\sin\varepsilon]$$

$$= [x\cos\theta - y\sin\theta]$$

$$w_2 = r\sin(\theta + \varepsilon) = r[\sin\theta\cos\varepsilon + \cos\theta\sin\varepsilon]$$

$$= [x\sin\theta + y\cos\theta]$$

$$w_1 = r\cos\theta\cos\varepsilon - r\sin\theta\sin\varepsilon = x\cos\theta - y\sin\theta$$

$$w_2 = r\sin\theta\cos\varepsilon + r\cos\theta\sin\varepsilon = x\sin\theta + y\cos\theta$$

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Rotation

The standard transposition matrix for a rotation of θ is $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Rotate $\langle 2;3 \rangle$ through $\frac{\pi}{6}$.

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} \cos\frac{\pi}{6} & -\sin\frac{\pi}{6} \\ \sin\frac{\pi}{6} & \cos\frac{\pi}{6} \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{3} - \frac{3}{2} \\ 1 + \frac{3\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} \frac{2\sqrt{3} - 3}{2} \\ \frac{2 + 3\sqrt{3}}{2} \end{bmatrix}$$

Transpose <1;1> by first rotating through $\frac{\pi}{4}$ and then reflecting through the y-axis.

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\frac{\pi}{4} & -\sin\frac{\pi}{4} \\ \sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ \frac{2}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{2}{\sqrt{2}} \end{bmatrix} = < 0; \frac{2}{\sqrt{2}} >$$